



A Hierarchical Online Monitoring and Multi-Model Spatiotemporal Tracing Framework for Abnormal Discharge in Urban Sewer Networks: A Case Study in Dongguan, China

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Abstract: Abnormal industrial wastewater discharge into urban sewer networks can disrupt the operation of wastewater treatment plants (WWTPs) and increase pollution risks to receiving water bodies. However, large-scale deployment of total phosphorus (TP) chemical monitoring at numerous sewer nodes is costly and difficult to maintain for high-frequency data acquisition, limiting rapid anomaly identification and source tracing. This study develops a hierarchical online monitoring and spatiotemporal tracing framework using the service area of the Xiani WWTP in Dongguan, China, as a case study. The study area was divided into six monitoring zones and 169 drainage blocks, with 57 online monitoring stations deployed at trunk and branch/secondary pipelines. Class-A stations directly monitored TP, while Class-B stations continuously monitored electrical conductivity (EC), pH, and liquid level. A “plant–network–zone–block–source” association framework was established by integrating pipe network topology, flow direction, and estimated travel time. Random forest-based contribution analysis, Fréchet distance trajectory matching, and Bayesian-Markov Chain Monte Carlo (MCMC) inference were combined to progressively rank candidate nodes, drainage blocks, and discharge sources. Results showed a moderate positive correlation between TP and EC (Spearman’s $\rho = 0.50$, $p < 0.05$, $n = 116$), indicating that EC can serve as an auxiliary indicator for screening phosphorus-containing industrial wastewater inputs but cannot replace direct TP measurement. In two verified abnormal discharge events, EC changes at Station 19B and Station 17B preceded TP anomalies at the WWTP influent by approximately 4 h and 6 h, respectively, consistent with estimated flow travel times. Field investigations further identified abnormal industrial wastewater discharge clues in the corresponding areas. Following targeted inspections, enterprise rectification, and enforcement actions, TP concentration at the A-Dapu Station decreased by 50%. The study demonstrates that the proposed hierarchical monitoring and multi-model tracing framework can progressively narrow anomaly investigation from the WWTP service area to candidate pipelines, drainage blocks, and discharge sources, supporting abnormal discharge screening and digital drainage management in industrially dense urban areas.

Keywords: Urban sewer network; Abnormal discharge; Online water quality monitoring; Multi-model spatiotemporal tracing; Digital drainage governance; Total phosphorus

1 Introduction

In recent years, the focus of urban sewage treatment in China has gradually shifted from the construction of new facilities to the improvement of treatment efficiency, the optimization of pipe network systems, and the refinement of operation and management mechanisms. The 14th Five-Year Plan for Urban Sewage Treatment and Resource Utilization Development [1] proposes that sewage treatment in water environment-sensitive areas should basically meet the Class 1A discharge standard. The Three-Year Action Plan for Improving the Quality and Efficiency of Urban Sewage Treatment (2019–2021) [2] requires the establishment and improvement of a long-term mechanism for drainage management, the regulation of industrial enterprise drainage behavior, and the strengthening of supervision over excessive discharge and illegal discharge through multi-department collaborative law enforcement. These policy requirements indicate that urban water environment governance is no longer limited to the improvement of end-of-pipe

capacity of sewage treatment facilities, but has further extended to the operation of drainage pipe networks, pollution source identification, and abnormal discharge disposal.

Dongguan is an important modern manufacturing city in China and a key component of the manufacturing system of the Guangdong–Hong Kong–Macao Greater Bay Area [3]. Qingxi Town is a typical area with rapid industrialization and urbanization in Dongguan, and has formed a certain scale of optoelectronic communication industry cluster. With the increase in the number of industrial enterprises, different types of production wastewater continuously enter the urban drainage system, placing higher demands on the influent quality and operation load of wastewater treatment plants (WWTPs). In particular, excessive discharge, illegal discharge, or wastewater leakage from industrial enterprises may rapidly alter the influent quality of WWTPs and exert pressure on subsequent treatment processes and receiving water bodies [4, 5]. According to the Dongguan 2022 Water Pollution Prevention and Control Work Plan, the tributary section of Xieken Water in Qingxi Town needs to eliminate inferior Class V water quality. To achieve this governance goal, it is not only necessary to improve the treatment capacity of WWTPs, but also to strengthen the dynamic monitoring of abnormal discharge in sewage pipe networks, and timely identify pollution transmission paths and their possible sources.

With the development of Internet of Things (IoT), field sensing, and digital water technology, water quality monitoring has gradually shifted from low-frequency manual sampling to continuous data acquisition, anomaly identification, and networked management [6–8]. By collecting water quality data from different pipe network nodes and combining them with pipe network topology, pollutant transport processes, and field verification, the scope of investigation for abnormal discharge sources can be gradually narrowed [9–11]. Relevant studies have applied sensor networks, pipe network hydraulic models, and probabilistic analysis to abnormal discharge identification, illegal connection localization, and monitoring station optimization [10–13]. However, existing applications usually rely on a small number of fixed monitoring stations or periodic manual sampling, making it difficult to balance large-scale pipe network coverage with high temporal resolution. For urban pipe networks with a large number of nodes and dense distribution of industrial dischargers, directly configuring complete chemical monitoring equipment at all levels of pipe network nodes is also limited by equipment cost, installation space, and operation and maintenance conditions.

Abnormal discharge in the sewage pipe network of Qingxi Town is mainly characterized by elevated total phosphorus (TP) concentrations. According to the current standard analytical method in China, TP determination usually requires steps such as sample digestion, reagent reaction, and photometric analysis [14]. Compared with parameters such as pH, electrical conductivity (EC), and liquid level that can be continuously obtained through field probes, direct determination of TP places higher demands on sample processing, analytical equipment, and operation and maintenance. Although methods such as portable X-ray fluorescence have been used for the rapid determination of TP and heavy metals in wastewater and sludge, their application still involves sample preparation, instrument calibration, matrix effect control, and detection result verification, and cannot be simply equated with continuous in-situ monitoring under pipe network conditions [15]. Therefore, large-scale deployment of TP chemical analysis equipment at branch and secondary pipe nodes with large quantities, narrow spaces, and complex operating conditions is not only limited by equipment cost and installation space, but also requires continuous investment in reagents, maintenance, and quality control. Although manual sampling can obtain relatively reliable detection results, the sampling frequency and analysis timeliness are limited, making it difficult to timely capture abnormal discharge events that are short in duration or occur during non-working hours. How to utilize water quality data collected by monitoring stations at different levels, screen on-site measurable indicators that can assist in reflecting abnormal changes in TP, and trace the source of abnormal discharge by combining pipe network spatial relationships and pollutant transport time, is a practical problem that needs to be solved in the digital governance of urban sewage pipe networks.

In response to the above problems, this study takes the service area of the Xiani WWTP in Qingxi Town, Dongguan as the object, and constructs a hierarchical online monitoring and multi-model spatiotemporal tracing framework for abnormal discharge in urban sewage pipe networks. The study first sets up different types of monitoring stations according to the service scope of the WWTP, pipe network structure, and drainage block distribution, and analyzes the correlation between TP and high-frequency monitorable water quality indicators. Subsequently, the corresponding relationships among the WWTP, trunk and branch pipe networks, monitoring nodes, drainage blocks, and potential pollution sources are sorted out, and a “plant–network–zone–block–source” association database is established. On this basis, the physically reachable candidate paths are constrained by pipe network topology and estimated flow travel time. Random forest regression is used to analyze the relative contribution of candidate nodes, Fréchet distance is used to compare the similarity of abnormal trajectories between upstream and downstream, and Bayesian-Markov Chain Monte Carlo (MCMC) is used to fuse spatial, temporal, and concentration response evidence to form a priority ranking of candidate drainage blocks and discharge sources, which is finally confirmed through field inspections and water sampling. This paper aims to illustrate how hierarchical online monitoring and multi-model evidence fusion can be used for the identification and source management of abnormal discharge in urban sewage pipe networks, and summarizes the practical application conditions of this method in areas with dense industrial enterprises, providing a

reference for urban digital drainage governance under the constraints of monitoring cost and deployment space.

2 Study Area and Hierarchical Monitoring Design

2.1 Overview of the Study Area

The study area is located within the service area of the Xiani WWTP in Qingxi Town, Dongguan. The designed treatment capacity of Xiani WWTP is 65,000 m³/d. It adopts the A²/O micro-aeration oxidation ditch process, and the effluent complies with the Class 1A standard of the Discharge Standard of Pollutants for Municipal Wastewater Treatment Plants (GB 18918—2002) [16]. The designed influent TP concentration of the WWTP is 4 mg/L. However, affected by abnormal discharge of high-concentration industrial wastewater within the service area, the influent TP concentration has remained at a high level for a long time, approaching or reaching 10 mg/L during some periods. Its variation trend is shown in Figure 1.

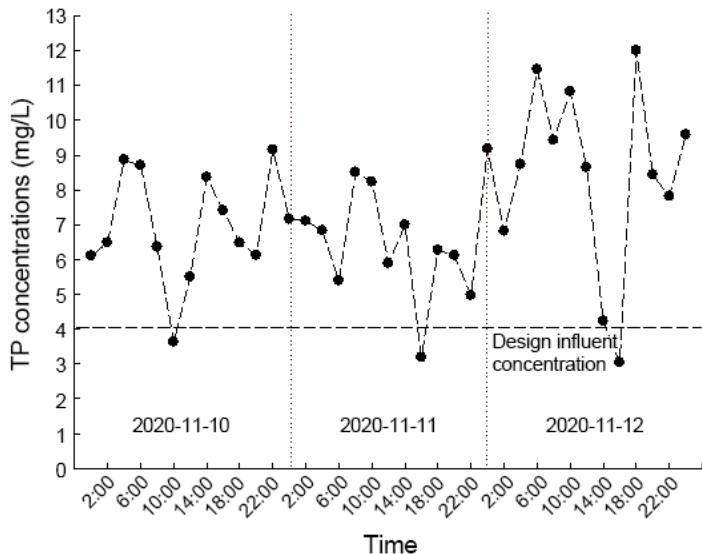


Figure 1. Variation trend of influent TP concentration of Xiani WWTP

Note: TP = total phosphorus; WWTP = wastewater treatment plant.

The continuous occurrence of abnormal influent water quality has caused obvious impacts on the normal operation of the WWTP. To avoid the impact of high-concentration industrial wastewater on the biochemical treatment system, the WWTP needs to reduce the influent flow rate, and the actual operation load rate has been maintained at about 80% for a long time. This operational constraint increases the pressure on regional wastewater collection and treatment systems and may further affect the protection of receiving water bodies.

Within the service area of the Xiani WWTP, there are 143 registered water-related enterprises in total, among which 74 are engaged in metal surface treatment and electronic processing and manufacturing, accounting for 52% of all water-related enterprises. The number of such enterprises is relatively large and their distribution is relatively scattered. The wastewater generated in their production process may contain phosphorus and heavy metal pollutants at high concentrations. Therefore, this area requires not only continuous monitoring of the influent water quality of the WWTP, but also the installation of monitoring stations at different levels of pipe network nodes to identify pollutant transmission paths and narrow the scope of investigation for abnormal discharge sources.

2.2 Hierarchical Monitoring Station Deployment

In June 2021, the Dongguan Ecological Environment Bureau launched a pilot project for intelligent supervision of water pollution prevention and control in Qingxi Town, to carry out refined and digital management of sewage pipe networks. According to the service scope, pipe network structure, sewage flow direction, and distribution of drainage enterprises of the Xiani WWTP, the study area was divided into 6 monitoring zones and 169 drainage blocks, and a total of 57 online water quality monitoring stations were set up at key nodes of trunk pipes and branch/secondary pipes. The monitoring system consists of Class-A stations and Class-B stations, which undertake the functions of zone water quality monitoring and branch/secondary pipe anomaly screening, respectively. Adopting monitoring nodes with different functions and deployment densities can balance monitoring coverage, time resolution, and equipment cost [12, 13]. The monitoring indicators, sampling frequency, deployment locations, and monitoring scope of each type of station are shown in Table 1.

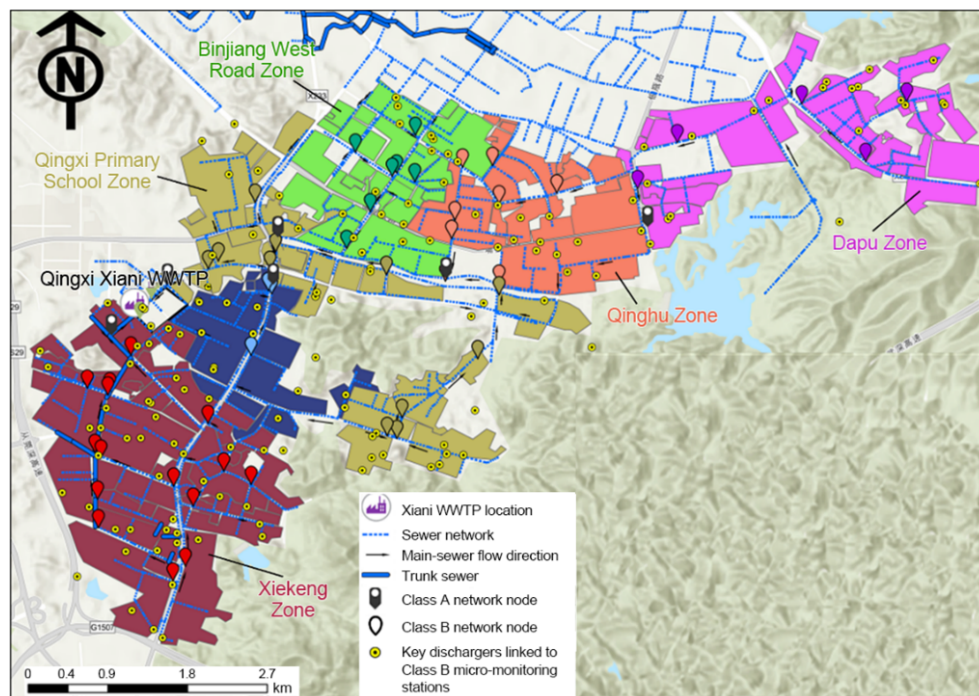
Table 1. Types and functions of online monitoring stations for pipe networks

Equipment Type	Monitoring Indicators	Sampling Frequency	Deployment Location	Monitoring Scope
Class-A station	Flow rate, TP (chemical method), pH, EC, liquid level	Once per hour	Trunk pipes	Average control of 20 km pipe network, 25 key drainage dischargers
Class-B station	pH, EC, liquid level	Once every 15 minutes	Branch/secondary pipe nodes	Average control of 2 km pipe network, 2–3 drainage blocks, 5–6 key drainage dischargers

Note: TP = total phosphorus; EC = electrical conductivity.

Class-A stations are mainly used to grasp the influent water quality of trunk pipes and each monitoring zone. A total of 6 Class-A stations using standard chemical analysis methods to monitor TP were set up on trunk pipes. Among them, the A-Xiekeng Station was set up on the west side pipe network; on the east side pipe network, according to the order from upstream to downstream, the A-Dapu, A-Qinghu, A-Binhe West Road, A-Tuqiao, and A-Qingxi Primary School stations were set up in sequence. The 6 Class-A stations correspond to 6 monitoring zones respectively, and are used to judge the water quality status and spatial differences of the water entering the trunk pipes from different zones.

Class-A stations use chemical methods to determine TP, and the equipment cost, space requirement, and operation and maintenance requirements are relatively high, making it difficult to deploy them on a large scale at branch/secondary pipe nodes. Therefore, this study further divided drainage units within each monitoring zone and deployed 51 Class-B online water quality monitoring stations in key branch/secondary pipe inspection wells. Class-B stations mainly use in-situ sensors to continuously monitor pH, EC, and liquid level, with a sampling frequency of once every 15 minutes. Each station controls an average of 2–3 drainage blocks and 5–6 key drainage enterprises. EC and liquid level sensors have the characteristics of fast response, low power consumption, and relatively convenient on-site deployment, and have been used for continuous monitoring and abnormal influent screening of urban drainage systems [17, 18]. Compared with TP chemical analysis equipment that requires reagent reactions, such sensors are smaller in size and convenient for distributed deployment in inspection wells, but cannot directly determine TP.

**Figure 2.** Distribution of pipe network zones and online water quality monitoring stations in the study area

Note: WWTP = wastewater treatment plant.

Based on the above differences, this study adopts a hierarchical deployment method of “direct monitoring on trunk pipes—characteristic indicator monitoring on branch/secondary pipes.” Class-A stations provide chemical monitoring

results of TP and other water quality indicators to judge the main zones involved in abnormal influent water; Class-B stations provide branch/secondary pipe monitoring data with wider coverage and higher time resolution for screening suspected abnormal nodes. On this basis, the relationship between TP and continuously monitorable indicators at Class-B stations is further analyzed to provide a data basis for the spatiotemporal tracing of abnormal discharge. The spatial distribution of the hierarchical monitoring stations and corresponding pipe network zones is presented in Figure 2.

3 Identification of Abnormal Discharge and Multi-Model Spatiotemporal Tracing Method

3.1 Screening of Water Quality Characteristic Indicators

Since Class-B stations cannot directly measure TP, this study needs to screen characteristic indicators related to TP variation from the water quality indicators that can be continuously monitored by Class-B stations, so as to establish the connection between the monitoring data of Class-A stations and Class-B stations, and provide a data basis for abnormal discharge identification in branch/secondary pipes, candidate path generation, and multi-model spatiotemporal tracing.

After analyzing the drainage characteristics of different types of enterprises in the study area, it was found that abnormal discharge of high-concentration phosphorus-containing industrial wastewater may be the main reason for the excessive TP in the influent of the Xiani WWTP. There are many metal surface treatment enterprises distributed in this area, and their production processes such as chemical polishing and phosphating generate phosphorus-containing wastewater. Metal surface phosphating treatment generates industrial wastewater containing phosphorus and possibly metal components such as iron, manganese, and nickel, and its ionic composition is significantly different from that of general domestic sewage [19]. The dissolved ion content in such wastewater is relatively high, and after being discharged into the sewage pipe network, it may cause obvious changes in EC. Therefore, this study takes EC as a candidate characteristic indicator for abnormal TP discharge, and uses the synchronous monitoring data of Class-A stations to test the correlation between the two.

According to the monitoring data of the 6 Class-A stations in August 2021, the TP pollution in the upstream pipe section where the Dapu Station is located was the most obvious, with its monthly average TP concentration being 3.3 mg/L. After the sewage is transported downstream along the trunk pipe, affected by the inflow and dilution of other incoming water, the TP concentration shows a general downward trend. The monthly average TP concentrations at the Qinghu Station, Binhe West Road Station, and Qingxi Primary School Station were 1.9, 1.7, and 1.6 mg/L, respectively. The spatial distribution of TP concentration in each monitoring zone is shown in Figure 3. There are 22 key drainage enterprises in the drainage area corresponding to the Dapu Station, of which 14 belong to the metal products and electronic equipment manufacturing industries, further indicating that this zone may be a key investigation area for phosphorus-containing industrial wastewater discharge.



Figure 3. Influent TP concentration of the Xiani WWTP and each monitoring zone

Note: TP = total phosphorus; WWTP = wastewater treatment plant.

To test whether EC can be used as an auxiliary indicator for identifying abnormal changes in TP, this study selected

the synchronous monitoring data of each Class-A station in August 2021 to conduct a correlation analysis between TP and other water quality indicators. Considering that the monitoring data may not meet the normal distribution, Spearman correlation analysis was used to test the relationship between TP and EC. The results showed a statistically significant moderate positive correlation between TP and EC (Spearman's $\rho = 0.50$, $p < 0.05$, $n = 116$). Among them, the changes in TP and EC at the Dapu Station, Qinghu Station, and Qingxi Primary School Station showed relatively obvious synchrony, and the specific variation trends are shown in Figure 4.

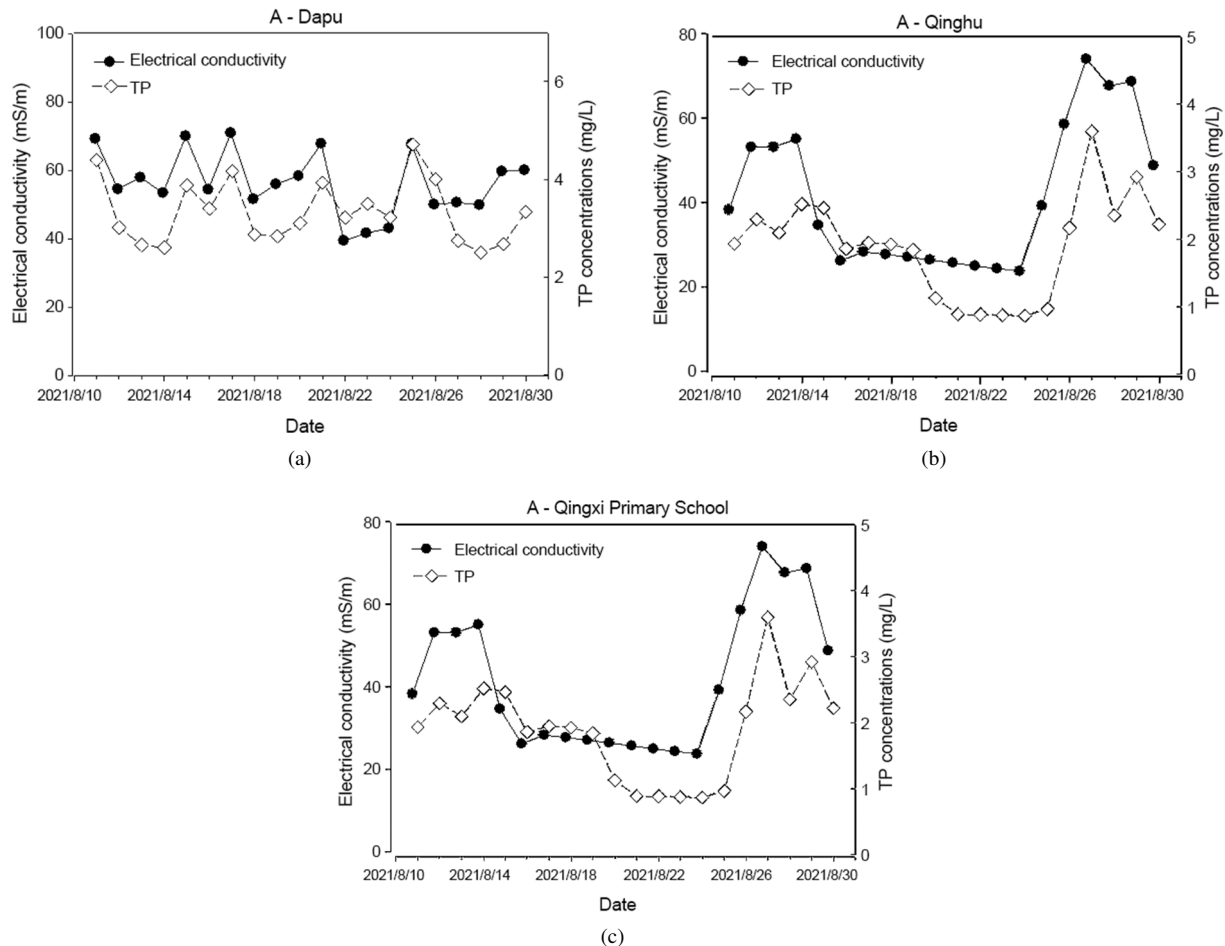


Figure 4. Variation trends of total phosphorus (TP) and electrical conductivity (EC) at different monitoring stations: (a) A-Dapu Station; (b) A-Qinghu Station; (c) A-Qingxi Primary School Station

The above results indicate that EC cannot replace the direct chemical determination of TP, but can be used as an auxiliary characteristic indicator for identifying abnormal discharge of phosphorus-containing wastewater in branch/secondary pipes. When the EC monitored by a Class-B station deviates significantly from the normal level, it can be combined with the types of upstream drainage enterprises, pipe network flow direction, and TP changes at Class-A stations to judge whether there is abnormal discharge of phosphorus-containing industrial wastewater. Existing studies have used EC for abnormal input identification and illegal discharge screening in urban drainage systems, but this indicator usually needs to be used in combination with water level, temperature, other water quality parameters, or field investigations [9, 10, 17].

3.2 Abnormal Discharge Identification and Multi-Model Spatiotemporal Tracing Process

This study takes the abnormal influent of the Xiani WWTP as the end response, sorts out the spatial and drainage relationships among the WWTP, trunk pipes, branch/secondary pipe monitoring nodes, monitoring zones, drainage blocks, and key drainage enterprises, and constructs a “plant–network–zone–block–source” directed association database. Among these, “plant” refers to the Xiani WWTP, “network” refers to the trunk pipe and branch/secondary pipe systems connected to the WWTP, “zone” and “block” correspond to the monitoring zones and the drainage blocks divided within them, respectively, and “source” refers to the key drainage enterprises or other discharge sources within the blocks that may generate abnormal discharge. The database records the monitoring station level, upstream

and downstream connection relationships, sewage flow direction, pipe segment length, flow velocity range, stations' covered blocks, distribution of key drainage dischargers, and the transmission path of sewage entering the WWTP, and is used to generate physically reachable candidate links and narrow the investigation scope step by step.

The tracing analysis uses the influent monitoring data of the WWTP, the data of Class-A stations on trunk pipes, the data of Class-B stations on branch/secondary pipes, and the IoT monitoring data at the discharge ends of some key enterprises. Data from different sources are sequentially subjected to outlier cleaning, format conversion, time resampling and synchronization, standardization, and spatial association, and organized according to monitoring time, station location, pipe network level, and drainage block. Data cleaning combines equipment operating status to identify sudden changes, drift, missing values, and technical anomalies caused by faults; time consistency processing is used to unify the data benchmarks of different sampling frequencies; standardization processing converts indicators of different dimensions and background levels into comparable relative anomaly sequences. Only data that pass quality verification enter the subsequent model analysis [20, 21].

When the influent TP concentration of the WWTP exceeds the set limit or significantly deviates from the normal level, the TP monitoring results of each Class-A station are first compared to determine the monitoring zone to which the abnormal influent originated; then the EC changes at Class-B stations within that zone are retrieved to screen the branch/secondary pipe nodes with abnormal peaks within the corresponding time range. When the hourly average EC of a Class-B station exceeds 100 mS/m and reaches more than 2 times the average EC of the previous 30 normal operation days at that station, it is marked as an abnormal candidate station. This threshold is only used to screen the pipe segments that need priority analysis, and is not used to directly identify polluting enterprises or estimate TP concentration.

For each abnormal candidate station, the actual transmission path from the station to the WWTP is determined according to the pipe network connection relationship. The estimated flow travel time for sewage to reach the WWTP from the candidate station is calculated according to the length of each pipe segment in the path and the corresponding sewage flow velocity:

$$T_j = \sum_{i=1}^k \frac{L_i}{v_i} \quad (1)$$

where, T_j is the estimated flow travel time for sewage to reach the WWTP from the j -th candidate station; k is the number of pipe segments included in the transmission path; L_i is the length of the i -th pipe segment; v_i is the sewage flow velocity corresponding to that pipe segment. The pipe segment length is determined according to the project pipe network data and monitoring station locations, and the sewage flow velocity is obtained from the pipe network flow velocity data mastered during the project operation period. For different pipe segments, the flow travel time is calculated and accumulated; when the sewage flow velocity has a certain variation range, the maximum and minimum flow velocities are used to calculate the estimated flow travel time range.

To represent the variation in flow velocity under phased operating conditions, let the directed reachable path from candidate node j to the WWTP be $\pi_j = (e_{j1}, e_{j2}, \dots, e_{jk_j})$, where L_{ji} denotes the length of the j -th pipe segment, and $v_{ji}^{\min}$ and $v_{ji}^{\max}$ represent the minimum and maximum flow velocities of this pipe segment, respectively. The lower and upper bounds of the estimated flow travel time along this path are calculated as:

$$\begin{aligned} T_j^- &= \sum_{i=1}^{k_j} \frac{L_{ji}}{v_{ji}^{\max}}, \\ T_j^+ &= \sum_{i=1}^{k_j} \frac{L_{ji}}{v_{ji}^{\min}} \end{aligned} \quad (2)$$

where, T_j^- and T_j^+ represent the shortest and longest estimated flow travel times from candidate node j to the WWTP, respectively, and k_j is the number of pipe segments included in the path. If the path exists and the direction of each pipe segment is consistent with the sewage flow direction, the topological reachability indicator χ_j^{topo} is defined as 1; otherwise, it is defined as 0. Pipe segment lengths are derived from the project pipe network data and the spatial locations of monitoring points, while the flow velocity ranges are obtained from phased pipe network operation data collected during the project period. Since dynamic hydraulic calibration results under continuous flow conditions were unavailable, these velocity ranges are used only as hydraulic constraints for estimating propagation time delays.

The transmission of pollutants in the pipe network is not solely determined by the straight-line distance between the monitoring station and the WWTP, but is jointly affected by pipe network connection relationships, upstream-downstream direction, pipe segment length, sewage flow velocity, and branch pipe confluence processes. Therefore, this study determines the actual transmission path from the candidate station to the WWTP based on the pipe network topology, rather than adopting the straight-line distance between the two points. For cases where multiple potential

transmission paths exist, the pipe segments that abnormal sewage may pass through are determined by combining the pipe network flow direction and upstream-downstream connection relationships, and the flow travel time of each pipe segment is calculated separately. Therefore, the velocity–distance calculation is used in this study to form a hydraulic propagation time constraint, rather than serving as the sole method for independent source determination.

The location of monitoring nodes and their coverage determine the spatial scale that tracing can achieve. In addition to the pipe network topology and flow travel time constraints, this study introduces random forest regression [22], Fréchet distance matching [23], and Bayesian-MCMC probabilistic inference [24] to form a multi-model evidence fusion framework that combines spatial, temporal, and concentration response information. The topological relationship is used to limit physically reachable candidate paths, and the flow travel time is used to constrain possible propagation time delays; random forest, Fréchet distance, and Bayesian-MCMC provide three types of evidence—relative contribution, abnormal trajectory matching, and posterior probability of candidate sources, respectively—which are then comprehensively ranked by the engineering platform. This framework draws on the application of sensor networks, machine learning, and probabilistic tracing methods in urban drainage anomaly identification [7, 11], but the model results are only used to narrow the scope and determine the priority of field investigation.

Subsequently, the observed time difference between the EC anomaly at the Class-B station and the TP anomaly in the influent of the WWTP is calculated:

$$\Delta t_j = t_{TP} - t_{EC,i} \quad (3)$$

where, Δt_j is the anomaly time difference between the j -th candidate station and the WWTP; t_{TP} is the time when the influent TP of the WWTP begins to rise significantly or reaches an abnormal peak; $t_{EC,j}$ is the time when the EC at the corresponding Class-B station begins to rise significantly or reaches an abnormal peak. Considering that the period from the beginning of the rise to the peak of a water quality indicator may last for some time, this study compares both the anomaly start time and the peak time to determine whether the observed time difference is basically consistent with the estimated flow travel time.

Combining pipe network topological reachability and propagation time-delay consistency, the preliminary spatiotemporal feasible candidate set is defined as:

$$C_{st} = \{j | \chi_j^{\text{topo}} = 1, T_j^- \leq \Delta t_j \leq T_j^+\} \quad (4)$$

where, C_{st} is the spatiotemporal feasible candidate set. This constraint is only used to exclude candidate nodes that are not connected in terms of pipe network direction or have obviously unreasonable propagation time delays, and is not equivalent to the final identification of pollution sources.

When the time difference between the EC anomaly at a Class-B station and the TP anomaly in the influent of the WWTP falls within the estimated flow travel time range, the monitoring data from the two locations are considered temporally consistent, and the Class-B station and its covered drainage blocks are retained in the preliminary candidate set. If multiple Class-B stations within the same zone satisfy the time matching condition, the anomaly magnitude of each station relative to the normal level is further compared, and the pipe network upstream-downstream relationship, drainage blocks, industry types of key enterprises, and monitoring results at the enterprise end are combined to form a preliminary priority; the final field investigation sequence is comprehensively determined based on the evidence fusion framework integrating the random forest, Fréchet distance, and Bayesian-MCMC described below.

The random forest regression model [22] is used to analyze the nonlinear relationship between candidate upstream monitoring characteristics and downstream abnormal responses, and to evaluate the relative contribution of different candidate nodes and drainage blocks to the plant-end anomaly accordingly. After completing data quality control, time synchronization, and flow travel time correction, the EC, pH, liquid level, and their abnormal variation characteristics from Class-B stations and enterprise discharge ends are used as model inputs, and the standardized concentration or abnormal intensity of TP in the influent of the downstream Class-A station or WWTP during the corresponding period is used as the model output.

The model builds multiple regression trees by repeatedly and randomly extracting historical monitoring records. The training records and monitoring indicator combinations used by different regression trees are not exactly the same, and each predicts the downstream abnormal response separately; the prediction results of all regression trees are averaged to obtain the ensemble prediction value of the random forest:

$$\hat{y}_t = \frac{1}{R} \sum_{r=1}^R f_r(x_t) \quad (5)$$

where, x_t is the candidate upstream monitoring feature vector corresponding to the downstream response at time t after quality control, time synchronization, and flow travel time correction; $f_r(x_t)$ is the predicted value of the downstream

abnormal response generated by the r -th regression tree; R is the total number of regression trees included in the random forest; y_t is the actual abnormal response of the downstream Class-A station or WWTP influent at time t ; $\hat{y}_t$ is the average of the prediction results of all regression trees.

The engineering platform generates a relative contribution score based on the influence of the monitoring characteristics related to each candidate station on the prediction results. The higher the score, the higher the degree of association between the candidate node and its covered drainage blocks and the downstream abnormal response. The random forest results are then combined with pipe network topology, flow travel time, Fréchet trajectory matching results, and Bayesian-MCMC posterior probability to determine the field investigation sequence.

The Fréchet distance is used to measure the overall similarity between the abnormal variation curves of a candidate node and a downstream monitoring point. During calculation, the two curves are allowed to be monotonically aligned along the time direction, and the minimum of the maximum point-to-point distance among all possible alignments is sought; the smaller the distance, the closer the rising, peak, and falling processes of the two curves are. Let P and Q be two standardized anomaly curves, and Γ be the set of allowable time alignment functions. Then the continuous Fréchet distance is defined as [23]:

$$d_F(P, Q) = \inf_{\alpha, \beta \in \Gamma} \max_{u \in [0, 1]} \|P(\alpha(u)) - Q(\beta(u))\|_2 \quad (6)$$

Considering that it takes some time for pollutants to propagate from the candidate node to the downstream, the standardized anomaly curve of candidate node j is shifted backward by a time translation τ ; Q is the standardized anomaly response curve of the downstream Class-A station or WWTP. Within the estimated flow travel time range of the candidate node, the minimum matching distance that the two curves can achieve is:

$$D_j^F = \min_{\tau \in [T_j^-, T_j^+]} d_F(P_j^\tau, Q) \quad (7)$$

In Eq. (7), τ is the time translation of the anomaly curve of the candidate node, the upper and lower limits of the interval are the shortest and longest estimated flow travel times from the candidate node to the downstream, and the left side of the equation is the minimum Fréchet distance that can be obtained within this time range. A smaller Fréchet distance indicates greater similarity between the anomaly trajectory of the candidate node and the downstream anomaly response, resulting in a higher tracing matching degree.

Bayesian-MCMC is used to synthesize evidence such as pipe network topology, flow travel time, and monitored concentration or anomaly magnitude, and calculate the posterior probability of each candidate discharge source. Let s_j denote the j -th candidate discharge source in the spatiotemporal feasible candidate set, θ denote the uncertain parameters in source strength and propagation process, and $\mathcal{D}$ denote the comprehensive evidence used by the model, then the posterior distribution is:

$$p(s_j, \theta | \mathcal{D}) \propto p(\mathcal{D} | s_j, \theta) p(s_j, \theta) \quad (8)$$

In Eq. (8), the first term on the right side is the likelihood of the candidate source and related parameters, and the second term is the prior distribution of observing the existing evidence given the candidate source s_j and parameters θ . MCMC repeatedly draws candidate sources and uncertain parameters from the above posterior distribution; if M valid samples are retained, the posterior probability estimate of candidate source j is:

$$\hat{p}_j = \frac{1}{M} \sum_{r=1}^M \mathbb{I}(s^{(r)} = s_j) \quad (9)$$

where, $s^{(r)}$ is the candidate source corresponding to the r -th valid sample, $\mathbb{I}(\cdot)$ is the indicator function, M is the total number of valid samples, and the left side is the posterior probability estimate of candidate source j . The larger this probability is, the higher the degree of support the candidate source receives from comprehensive evidence such as topology, time, and monitoring response, and the engineering platform ranks the candidate discharge sources accordingly.

At the discharge ends of some key drainage enterprises, IoT monitoring equipment was also installed in the project, and hierarchical anomaly prompts and alarm rules were configured in the online monitoring platform. When the influent TP of the WWTP exceeds the designed influent concentration or significantly deviates from the recent background level, the management personnel initiate a retrospective review of the pipe network monitoring data; when the EC of a single Class-B station reaches the above abnormal condition, the platform marks it as a suspected abnormal node; when the hourly average TP of a Class-A station exceeds 8 mg/L, and the EC of its upstream and downstream associated Class-B stations reaches the abnormal condition, the system issues a combined alarm. After an alarm is triggered, the platform first generates candidate paths based on the “plant–network–zone–block–source” topology and flow travel time, and then integrates the random forest relative contribution, Fréchet trajectory similarity, and

Bayesian-MCMC posterior results to form a priority list of candidate pipe segments, drainage blocks, and discharge sources. The management personnel then cross-check the monitoring data of the WWTP, Class-A stations, Class-B stations, and enterprise ends; environmental management and law enforcement personnel carry out on-site inspections and water sampling for key drainage dischargers, scattered wastewater discharges, and unlicensed discharge clues in the top-ranked areas, so as to confirm the abnormal discharge source and take rectification or supervision measures.

Therefore, the abnormal discharge tracing process of this study sequentially includes multi-source data preprocessing, abnormal identification of WWTP influent, zone positioning by Class-A stations, screening of abnormal nodes by Class-B stations, candidate path generation under topology and flow travel time constraints, random forest contribution analysis, Fréchet trajectory similarity matching, Bayesian-MCMC posterior inference, model evidence integration and candidate source ranking, as well as field verification and water sampling confirmation. The velocity–distance calculation is only a hydraulic constraint in the multi-model framework; the model output cannot directly determine the specific polluting enterprise, and the specific pollution source still needs to be judged by combining enterprise drainage characteristics, field inspections, and detection results. The complete abnormal discharge identification and multi-model spatiotemporal tracing process is shown in Figure 5.

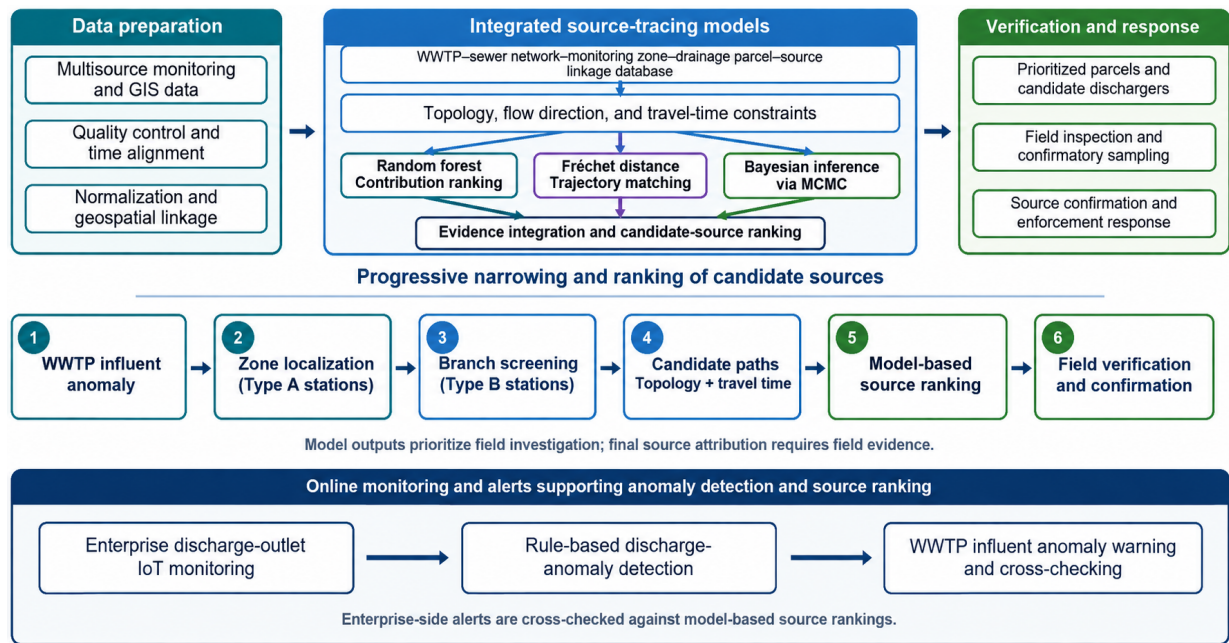


Figure 5. Multi-model workflow for abnormal-discharge identification and spatiotemporal source tracing in an urban sewer network

Note: GIS = geographic information system; MCMC = Markov chain Monte Carlo; WWTP = wastewater treatment plant.

4 Results of Abnormal Discharge Tracing and Governance Effectiveness

4.1 Tracing of Abnormal Discharge Events at Enterprise End

To examine the application of the hierarchical online monitoring and spatiotemporal tracing method in actual pipe network supervision, this study selected two abnormal events with complete monitoring records and field verification results for analysis. The overall engineering framework of the project platform includes abnormal identification of WWTP influent, retrospective review of pipe network monitoring data, screening of candidate nodes, multi-model candidate ranking, and field inspection and sampling.

The first abnormal event occurred on September 28, 2021. Starting from 12:00 on that day, the influent TP concentration of the Xiani WWTP began to rise and remained at about 5 mg/L, which was higher than the designed influent concentration of 4 mg/L. At the same time, the total concentration of five heavy metals—copper, zinc, manganese, nickel, and chromium—in the influent exceeded 5 mg/L. The synchronous increase in TP and multiple heavy metal concentrations indicated that this abnormal influent might be related to industrial wastewater input. The management personnel then initiated a retrospective review of the pipe network monitoring data and cross-checked the same-period monitoring data of upstream Class-A and Class-B stations.

Based on the monitoring data of trunk pipes and branch/secondary pipes and the “plant–network–zone–block–source” association relationship, nodes that did not meet the spatial connectivity and propagation time delay conditions were first excluded, and Station 19B was retained as a candidate node with a high degree of temporal and spatial association

with this event. Starting from 8:00 on September 28, the EC at this station increased from about 330 mS/m to about 1,400 mS/m, which was about 4 h earlier than the time when the influent TP of the WWTP began to rise. According to the estimation of the pipe network distance and sewage flow velocity from this station to the WWTP, it takes about 4 h for sewage to be transported from the pipe segment where Station 19B is located to the WWTP. The time difference of the monitoring data was basically consistent with the estimated flow travel time. Therefore, the pipe segment where Station 19B is located and its corresponding drainage block were listed as priority investigation areas.

On the night of September 30, the EC at Station 19B rose again to about 1,600 mS/m. The hourly average exceeded 100 mS/m and reached more than 2 times the normal daily average of the station, meeting the EC anomaly determination condition of Class-B stations. The online platform therefore marked the station as a suspected abnormal node. Combining the plant-end anomaly on September 28, the monitoring data of Station 19B, the pipe network flow direction, and the estimated flow travel time, the management personnel continued to list Station 19B and its controlled blocks as priority inspection targets. The changes in the influent of the WWTP and the monitoring indicators of Station 19B are shown in Figure 6a.

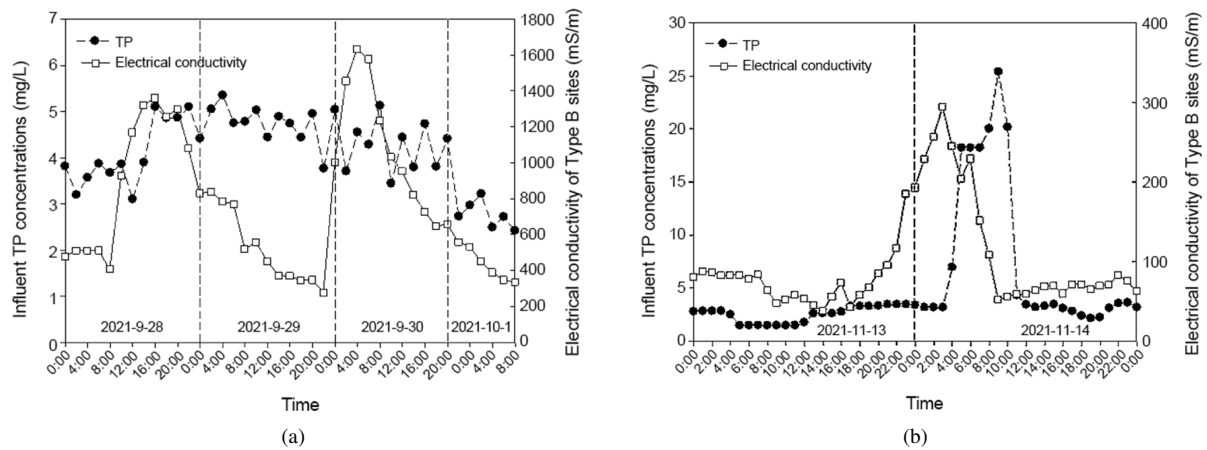


Figure 6. Water quality variations during two abnormal discharge events: (a) temporal relationship between the EC anomaly at Station 19B and influent TP variation at the Xiani WWTP; (b) temporal relationship between EC variation at Station 17B and the influent TP anomaly at the Xiani WWTP

Note: EC = electrical conductivity; TP = total phosphorus; WWTP = wastewater treatment plant.

Environmental law enforcement personnel subsequently carried out on-site investigation in the area controlled by Station 19B and found that an anode oxidation enterprise was suspected of abnormal discharge. The test results showed that the TP concentration of the concentrated reuse water within the enterprise was 36 mg/L, and there was a possibility of wastewater leakage. The enterprise was subsequently ordered to suspend production for rectification. After the disposal, the EC of Station 19B and the influent TP concentration of the WWTP both began to decline; by October 1, the influent TP concentration of the WWTP had dropped to below 4 mg/L, recovering to the designed influent concentration range.

The second abnormal event occurred on November 14, 2021. At 9:00 on that day, the influent TP concentration of the Xiani WWTP increased from the daily level of about 3 mg/L to 25.37 mg/L, triggering an influent abnormal alarm. The platform generated candidate paths based on the pipe network topology, abnormal node location, and estimated flow travel time, and Station 17B was retained as a candidate node with a high degree of temporal association with this event. The EC at this station began to rise at 0:00 on November 14 and reached a peak of about 300 mS/m at 3:00, which was significantly higher than the background level of 60–70 mS/m at the station on normal days. The variation process is shown in Figure 6b.

The time difference between the EC peak at Station 17B and the abnormal influent TP of the WWTP was about 6 h, which fell within the estimated flow travel time range of 4–6 h from the station to the WWTP. If calculated from 0:00 when the EC began to rise, the difference was about 9 h, exceeding the estimated flow travel time, indicating that the abnormal input might have lasted for some time. Therefore, for this event, the EC peak was more suitable as a reference for pollution propagation time matching than the time when the rise began. Combining the pipe network flow direction and the station's control scope, the area where Station 17B is located was listed as a priority field investigation area.

Law enforcement personnel subsequently carried out on-site inspections and sampling, and measured a chemical oxygen demand (COD) concentration of 2,800 mg/L and a TP concentration of 13 mg/L downstream of the discharge outlet of a hardware manufacturing enterprise in the area controlled by Station 17B. The on-site test results indicated

that the drainage of this enterprise might be an important source of the abnormal water quality in the pipe network. The enterprise was subsequently ordered to rectify and upgrade its wastewater treatment facilities.

The two events showed that there was a temporal association between the EC change at Class-B stations and the abnormal influent TP of the WWTP that could be used for tracing screening. For the 19B event, the time difference between the EC beginning to rise and the plant-end TP beginning to rise was about 4 h; for the 17B event, the time difference between the EC peak and the plant-end TP anomaly was about 6 h. The above time differences were basically consistent with the estimated flow travel times of the corresponding pipe segments. Combining the pipe network topology, station control scope, and field verification, the investigation scope could be narrowed down to candidate pipe segments and drainage blocks, and clues of abnormal industrial wastewater discharge in the corresponding areas could be found. Random forest, Fréchet distance, and Bayesian-MCMC belong to the candidate ranking module in the overall engineering framework, but since the two events did not retain reviewable numerical outputs of each model, it is not possible to judge the contribution of each model to the selection of specific stations or verify the performance of the combined model. The final identification of pollution sources still needs to be combined with enterprise production and drainage characteristics, field inspections, and water quality test results.

4.2 Zonal Governance and Influent Water Quality Changes

According to the spatial distribution results of TP in each monitoring zone, the TP pollution in the zone where the A-Dapu Station is located was relatively prominent. The monthly average TP concentration recorded by this station was higher than that of other trunk pipe monitoring stations, so it was listed as a key governance area. Relevant departments carried out investigations on the scattered pollution-related enterprises in the Dapu zone based on the monitoring results, organized three special law enforcement actions successively, and strengthened the daily supervision of key drainage enterprises. After taking the above measures, the TP concentration recorded by the A-Dapu Station decreased by 50%.

The pilot project for intelligent supervision of water pollution prevention and control in Qingxi Town was launched in June 2021. During the project operation period, no obvious changes occurred in the influent flow rate and COD concentration of the Xiani WWTP, while the influent TP concentration showed a general downward trend. In August 2021, the monthly average influent TP concentration of the WWTP dropped to 2.5 mg/L, which was the lowest value in the monitoring records at that time, a decrease of 42% compared with the same period of the previous year and a decrease of 25% compared with the previous month. After that, the year-on-year decrease in influent TP concentration exceeded 30% in each month, and the changes in different years are shown in Figure 7.

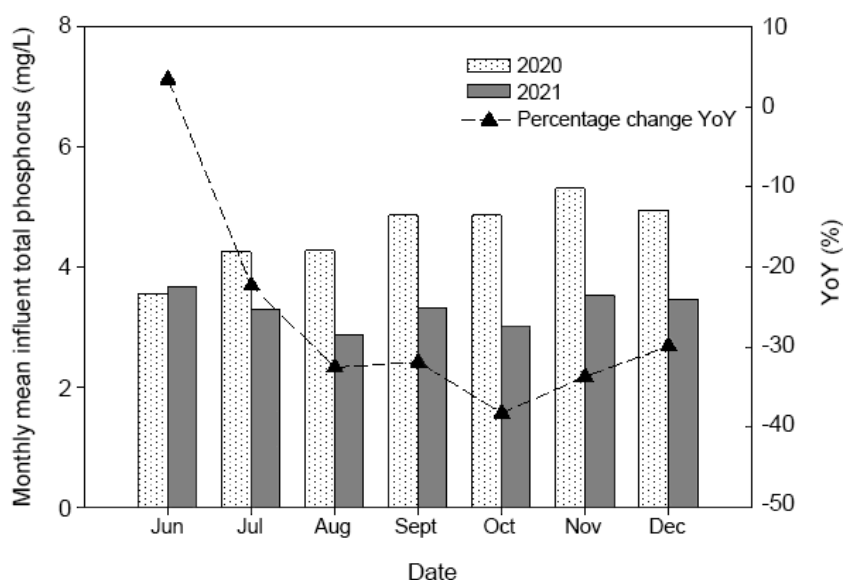


Figure 7. Comparison of influent TP concentration changes of the Xiani WWTP in different year

Note: TP = total phosphorus; WWTP = wastewater treatment plant; YoY = Year-over-Year.

These changes indicate that, under the condition of relatively stable influent flow rate and COD concentration, the abnormal area location information provided by the hierarchical online monitoring provided a basis for enterprise investigation, law enforcement inspection, and subsequent supervision. By combining monitoring results with on-site management measures, the management department can gradually narrow the investigation scope from the entire service area of the WWTP to specific zones, branch/secondary pipe nodes, and drainage blocks, thereby reducing the time and personnel investment required for untargeted investigation.

However, since multiple measures such as special law enforcement, enterprise rectification, and daily supervision were implemented during the same period, and this study did not set up a control area without this monitoring method, the decrease in TP concentration cannot be completely attributed to the online monitoring system. More accurately, hierarchical monitoring and spatiotemporal tracing provided location and decision-making basis for governance measures, and the water quality improvement was the result of the combined effect of monitoring and early warning, field law enforcement, and enterprise rectification.

5 Discussion

5.1 Role and Result Interpretation of the Hierarchical Monitoring Framework

The hierarchical online monitoring system constructed in this study utilizes the functional differences between different types of monitoring stations to narrow the investigation scope of abnormal discharge step by step. Class-A stations are set at trunk pipe nodes and use chemical methods to monitor TP, which is used to identify the main zones involved in abnormal influent; Class-B stations are set at branch/secondary pipe nodes and monitor indicators that are easy to measure continuously in situ, such as EC, at high frequency, to further screen suspected abnormal pipe segments. On this basis, the “plant–network–zone–block–source” association relationship connects the WWTP, trunk pipes, branch/secondary pipes, drainage blocks, and key drainage enterprises according to the pipe network topology, so that the abnormal changes of stations at different levels can correspond to specific spatial management units. Compared with relying solely on the influent data of the WWTP for post-event investigation, this method can gradually narrow the investigation scope from the entire catchment area to specific branch/secondary pipes and their corresponding drainage blocks.

Two abnormal events verified by field inspections preliminarily illustrated the feasibility of the joint screening of hierarchical monitoring, pipe network topology, and flow travel time. For the 19B event, the time difference between the EC beginning to rise and the plant-end TP beginning to rise was about 4 h; for the 17B event, the time difference between the EC peak and the plant-end TP anomaly was about 6 h, both of which were basically consistent with the estimated flow travel times of the corresponding pipe segments. These spatiotemporal evidences that can be directly reviewed provided a basis for candidate path screening. In the overall engineering framework, random forest, Fréchet distance, and Bayesian-MCMC are configured as candidate ranking modules, but the existing event archives did not retain their scores for each candidate, so the two cases cannot independently prove the incremental effect of these modules or the performance of the combined model. For subsequent events with complete records, it can be further tested whether the complete abnormal trajectory provides more discriminative information than a single start time or peak time.

The above cases indicate that pipe network topology and estimated flow travel time can eliminate candidate paths that are unreasonable in space or propagation time delay, but cannot complete source identification alone. In terms of method function, random forest can characterize the nonlinear relationship between multi-station monitoring characteristics and downstream anomalies, Fréchet distance can compare the overall shape of abnormal processes, and Bayesian-MCMC can estimate the posterior probability of candidate sources under uncertain conditions; the three types of outputs and hydraulic constraints together constitute the candidate ranking evidence. Since this study did not retain the model numerical outputs of the two events, these functions cannot be interpreted as having been quantitatively verified by the above cases. Domestic sewage fluctuations, other high ionic strength wastewater, branch pipe confluence, dilution and deposition, and sensor drift may all affect model inputs, so the final verification still needs to be completed through enterprise production and drainage characteristics, field inspections, and discharge outlet water quality sampling [10, 17, 20].

Existing studies have used multi-parameter monitoring, low-cost sensor networks, and hydraulic models or probabilistic inference methods to identify abnormal inputs in urban drainage pipe networks [9–12, 17]. The common point between this study and these methods is the use of continuous monitoring data to identify abnormal changes that deviate from the background state; the difference first lies in addressing the problem that TP is difficult to directly measure at high frequency at a large number of branch/secondary pipe nodes in areas with dense industrial enterprises, by adopting a hierarchical monitoring path of “direct TP measurement on trunk pipes—EC monitoring on branch/secondary pipes—field sampling confirmation.” Secondly, this monitoring path is combined with the “plant–network–zone–block–source” relationship database, pipe network topology and flow travel time constraints, random forest relative contribution, Fréchet trajectory matching, and Bayesian-MCMC candidate ranking. EC is only used in this framework to identify abnormal ionic inputs and screen candidate pipe segments, and is not used as a substitute measurement for TP concentration; this combination of hierarchical monitoring and multi-model evidence fusion can avoid misunderstanding the statistical correlation between EC and TP as a stable quantitative conversion relationship.

Overall, the field verification results of the two events provide preliminary evidence for the practical application of this method, but the existing number of cases is not enough to comprehensively evaluate the location accuracy, stability, and cross-event generalization ability of the combined model of random forest, Fréchet distance, and

Bayesian-MCMC. At present, the true positives, false positives, and false negatives of all alarm events have not been systematically counted, and training-validation split, model ablation, and parameter sensitivity analysis based on a unified ground truth definition are also lacking. Therefore, at this stage, it is more appropriate to define this method as a candidate source screening and investigation ranking tool for management applications, rather than an automatic diagnosis model that can independently confirm pollution sources.

5.2 Management Implications for Urban Drainage Governance

Traditional abnormal discharge investigations are often launched after excessive influent is detected at the WWTP. Since the investigation scope involves a large number of pipe segments and drainage enterprises, management personnel need to inspect section by section and carry out multi-point sampling, which is not only time-consuming but also likely to miss discharge events of short duration. The “plant–network–zone–block–source” association relationship established in this study incorporates the WWTP, trunk pipes, branch/secondary pipes, drainage blocks, and key drainage enterprises into a single investigation path, enabling water quality monitoring data to correspond to specific spatial management units. The management department can first determine the abnormal zone based on Class-A stations, then identify suspected branch/secondary pipes based on the water quality changes at Class-B stations, and finally focus the field investigation on the corresponding drainage blocks and key enterprises.

The role of the continuous monitoring system is not limited to recording water quality changes; it also lies in transforming the information scattered across the WWTP, trunk pipes, branch/secondary pipes, and enterprise discharge ends into evidence that can be used for model calculation, investigation ranking, and management decision-making. Existing studies have shown that real-time monitoring, data integration, and digital analysis can provide information support for the operational status judgment and abnormal event response of urban sewage systems [25, 26]. In this study, after the online platform issued an anomaly alert, the management personnel first checked the data quality of stations at different levels, then combined the pipe network topology, flow travel time, random forest relative contribution, Fréchet trajectory similarity, and Bayesian-MCMC posterior ranking to determine the priority investigation areas, followed by on-site inspections and water sampling by law enforcement personnel. Therefore, the system connects the links of anomaly discovery, model ranking, field verification, and enterprise rectification, rather than replacing the management and law enforcement personnel in making final judgments.

For industrial cities with a large number of water-related enterprises but limited on-site law enforcement capacity, hierarchical monitoring can help the management department determine the inspection sequence and reduce pipe network patrols and enterprise investigations that lack clear targets. During the project operation period, relevant departments carried out three special law enforcement actions in the Dapu zone based on the monitoring results, and conducted on-site inspections and rectifications on the enterprises involved in the two abnormal events. These practices indicate that monitoring information can only be transformed into actual governance actions when combined with on-site sampling, enterprise supervision, and subsequent rectification. Compared with simply increasing monitoring equipment, establishing clear processes for data verification, alarm handling, and inter-department collaboration is more important for the long-term operation of the system.

The influent TP concentration of the WWTP generally decreased after the project implementation, but this change cannot be completely attributed to the monitoring system. Online monitoring itself does not directly reduce pollution discharge; its role is to provide information for anomaly discovery, location screening, and law enforcement decision-making. The actual water quality changes are also affected by factors such as special law enforcement, enterprise production suspension and rectification, wastewater treatment facility upgrades, and daily supervision. Therefore, the results of this study reflect the comprehensive effect of the “monitoring–early warning–investigation–rectification” governance chain, rather than the independent effect of a single technical device.

5.3 Applicable Conditions and Promotion Value of the Method

The framework of this study is mainly applicable to areas where industrial dischargers are relatively concentrated, the hierarchical relationship of the drainage pipe network is relatively clear, the online monitoring coverage reaches a certain density, and abnormal wastewater can cause significant changes in characteristic water quality indicators. When applying it in other cities, it is necessary to master the service scope of the WWTP, the connection relationships of trunk and branch pipes, the main drainage directions, the distribution of key enterprises, and historical abnormal events, and establish the correspondence between monitoring stations and drainage blocks accordingly. Incomplete basic data of the pipe network, severe rain-sewage mixing, frequent changes in water flow direction, insufficient training samples, or uneven coverage of enterprise-end monitoring will all reduce the discrimination ability of candidate path constraints and model posterior ranking.

The candidate characteristic indicators also need to be re-screened according to the local industrial structure and main pollutant types. EC has a certain identification effect in the study area because there are many metal surface treatment enterprises locally, and phosphorus-containing wastewater is usually accompanied by changes in dissolved ion concentration. In areas dominated by food processing, organic chemical, or other industries, EC may not be the

most suitable indicator, and it may be necessary to combine turbidity, redox potential, organic matter indicators, or other on-site measurable parameters. It can be seen that the overall idea of hierarchical monitoring has a certain degree of transferability, but the specific monitoring indicators, alarm thresholds, and station deployment schemes cannot be directly copied, and need to be calibrated according to local drainage characteristics.

The station deployment density will also affect the tracing scope. The fewer drainage blocks and enterprises covered by a Class-B station, the smaller the investigation scope that can be determined after an anomaly occurs, but the equipment construction and maintenance costs will also increase accordingly. The site selection of sewage pipe network monitoring equipment is essentially a trade-off among pollution source identification capability, network coverage, and construction cost [12, 13]. When deploying the system, urban management departments need to comprehensively consider the distribution of key pollution sources, pipe network structure, abnormal discharge risks, and supervision resources, rather than simply pursuing the number of monitoring stations. For pipe segments with high pollution risks and areas with concentrated enterprises, the station density can be increased; for areas with a single drainage source or low risk, a lower-density monitoring layout can be adopted.

5.4 Research Limitations and Future Work

This study still has several limitations. First, only a moderate correlation was observed between TP and EC. The existing analysis is mainly based on 116 sets of synchronous monitoring data obtained in August 2021, and has not yet fully tested whether this relationship is stable under different seasons, rainfall conditions, and enterprise production cycles. Rainfall infiltration, changes in the proportion of domestic sewage, and the inflow of different types of industrial wastewater may all alter the background EC of the pipe network and its relationship with TP. High-frequency water quality monitoring can reveal short-term changes that are difficult to capture with low-frequency sampling [8], but its reliable application depends on reasonable monitoring design, continuous sensor maintenance, standardized data quality control, and appropriate background state identification [20, 21, 27]. Future research needs to expand the monitoring period, analyze the relationship between the two indicators at different stations, in different seasons, and under different hydrological conditions, and establish dynamic background ranges based on the long-term monitoring data of each station, rather than uniformly adopting the same fixed threshold across all stations and periods.

Second, this paper focuses on analyzing two abnormal discharge events with field verification results. Although both events presented a relatively consistent process of “monitoring anomaly—scope screening—time matching—field verification,” the sample size is still limited, which is insufficient to evaluate the overall identification performance of this method under long-term operation conditions. Existing studies have also not fully counted all alarm events during the project operation period, including anomalies whose sources could not be determined, false alarms, and discharge events that may not have been identified by the system. In the future, all alarm records during the study period should be compiled and classified into categories such as correct identification, false alarm, missed detection, and unconfirmed based on field verification results, so as to calculate the location accuracy, false alarm rate, missed detection rate, and average early warning lead time.

Third, there are uncertainties in each component of the multi-model tracing. Flow travel time is affected by flow rate, pipe diameter, slope, pump station operation, rainfall infiltration, and branch pipe confluence; adopting fixed or phased flow velocities may lead to propagation time delay deviations. The relative contribution of random forest is affected by the representativeness of training samples, feature correlation, and monitoring coverage, and cannot be directly interpreted as a causal discharge share. Fréchet distance is sensitive to data standardization, event window, and allowable time shift range. The posterior ranking of Bayesian-MCMC depends on the prior, likelihood, chain length, and convergence diagnosis. In the future, continuous flow and liquid level data should be used to dynamically update the flow travel time range, and model hyperparameters, candidate-by-candidate outputs, training-validation split, Fréchet preprocessing and event window, MCMC sampling and convergence diagnosis, and evidence fusion rules should be fully archived before model calibration, ablation analysis, and uncertainty evaluation are carried out through field-confirmed events.

Finally, this study did not set up a control area without the hierarchical monitoring framework, nor did it quantitatively compare the complete multi-model framework with traditional manual investigation, single-station threshold alarm, flow travel time matching only, or other hydraulic-probabilistic methods under a unified event ground truth and evaluation indicators. Future research should separately evaluate the incremental contribution of topological constraints, random forest, Fréchet distance, and Bayesian-MCMC to the narrowing of candidate scope and ranking stability, and record identification accuracy, investigation time, number of field samples, personnel input, and operation and maintenance costs. On this basis, service areas of WWTPs with different industrial structures and pipe network conditions can be selected for external validation to judge the cross-regional applicability of the model.

6 Conclusions

Aiming at the problems that abnormal influent TP of WWTPs in areas with dense industrial enterprises is difficult to locate in a timely manner, and it is difficult to directly carry out high-frequency TP monitoring on a large scale at

branch/secondary pipe nodes, this study took the catchment area of the Xiani WWTP in Qingxi Town, Dongguan City as the research object, and constructed a multi-model spatiotemporal tracing method integrating hierarchical online monitoring, “plant–network–zone–block–source” association relationship, pipe network topology and flow travel time constraints, random forest contribution analysis, Fréchet trajectory matching, and Bayesian-MCMC posterior inference. The overall engineering framework organizes direct TP monitoring on trunk pipes, high-frequency screening of EC on branch/secondary pipes, model candidate ranking, and on-site verification at enterprise ends into a continuous investigation path; the two actual abnormal events were used to analyze the directly reviewable processes of hierarchical monitoring, topology–flow travel time screening, and field verification, and are not used as verification of the combined model performance. The main conclusions are as follows.

(1) A hierarchical online monitoring network consisting of 6 Class-A stations and 51 Class-B stations was established in the study area. Class-A stations are set at trunk pipe nodes to directly monitor TP; Class-B stations are set at branch/secondary pipe nodes to monitor EC, pH, and liquid level at high frequency. The “plant–network–zone–block–source” association relationship further incorporates the WWTP, trunk pipes, branch/secondary pipes, drainage blocks, and key drainage enterprises into a single spatial investigation path, forming a step-by-step tracing process from plant-end anomaly discovery, zone identification, branch/secondary pipe screening, to enterprise field verification.

(2) The combined monitoring data of the 6 Class-A stations during the study period showed a moderate positive correlation between TP concentration and EC (Spearman’s $\rho = 0.50$, $p < 0.05$, $n = 116$). Combined with the wastewater characteristics of metal surface treatment enterprises in the study area, EC can serve as an auxiliary identification indicator for abnormal input of phosphorus-containing industrial wastewater, but cannot be used to directly determine or quantitatively predict TP concentration. This relationship may be affected by enterprise type, wastewater composition, pipe network dilution process, and station background differences, so it should not be interpreted as a fixed TP–EC conversion relationship.

(3) Two abnormal events verified by field inspections showed that EC anomalies at Class-B stations can provide clues for candidate pipe segment screening. Among them, the time difference between the EC peak at Station 17B and the abnormal influent TP of the WWTP was about 6 h, which fell within the estimated flow travel time range of 4–6 h for that pipe segment. Pipe network topology and flow travel time can be used to limit feasible propagation paths; random forest, Fréchet distance, and Bayesian-MCMC are used for candidate ranking in the overall engineering framework, but the two cases did not retain reviewable model-by-model outputs, so the location performance of the combined model cannot be verified from them. Specific pollution sources still need to be confirmed through field inspections and water sampling.

(4) During the project implementation period, the monthly average influent TP concentration of the Xiani WWTP generally decreased, and the TP concentration at the corresponding monitoring station also decreased after the special law enforcement in the Dapu zone. The above changes reflect the comprehensive effect of the governance chain composed of online monitoring, abnormal early warning, field investigation, enterprise rectification, and special law enforcement, and cannot be completely attributed to the online monitoring system itself. The application value of this method lies in transforming the monitoring information scattered across the WWTP, trunk pipes, and branch/secondary pipes into clear investigation sequences and spatial clues, reducing untargeted pipe network patrols, and providing a basis for drainage supervision, field law enforcement, and limited management resource allocation in areas with dense industrial enterprises.

(5) Compared with simply increasing TP monitoring equipment, relying solely on WWTP influent data for post-event investigation, or only performing flow travel time back-calculation, this study utilizes the functional complementarity of different levels of sensors and multi-model evidence to realize a continuous management process of “plant-end anomaly identification—hierarchical screening by Class-A/B stations—topology and flow travel time constraints—random forest and Fréchet analysis—Bayesian-MCMC candidate ranking—enterprise field verification.” This method transforms velocity–distance calculation from a single tracing method into a hydraulic constraint in the multi-model framework, providing a more complete implementation idea for the supervision of abnormal discharge in urban sewage pipe networks under the condition of limited monitoring resources.

Author Contributions

Conceptualization, W.H.G.; methodology, Q.Y.P. and Q.L.; software, Q.Y.P. and Q.L.; validation, Q.Y.P. and Q.L.; formal analysis, W.H.G.; investigation, Q.Y.P. and Q.L.; resources, Q.Y.P. and Q.L.; data curation, Q.Y.P. and Q.L.; writing—original draft preparation, W.H.G.; writing—review and editing, W.H.G.; visualization, W.H.G.; supervision, Q.Y.P. and Q.L.; project administration, Q.Y.P. and Q.L. All authors have read and agreed to the published version of the manuscript.

Data Availability

The raw datasets generated and/or analyzed during this study contain proprietary information provided by Guangdong Huihang Tianwei Technology Co., Ltd., and are not publicly available due to commercial confidentiality and contractual restrictions. Data may be requested from the corresponding author, but any release is subject to prior written approval from Huihang Tianwei Technology Co., Ltd. and, where required, execution of a data-sharing or non-disclosure agreement.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors declare that generative artificial intelligence (AI) or AI assisted technologies were used solely for language polishing and improving readability during the preparation of this manuscript. Specifically, the GPT model was employed to refine grammar, phrasing, and overall textual clarity. No generative AI tool was used to generate or manipulate data, results, references, or to replace substantive intellectual contributions. The authors take full responsibility for the content, originality, and accuracy of this work, and confirm that all ethical and publishing standards have been met.

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